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1 April 1969

MILITARY SPECIFICATION

CONTAINER, MISSILE

MK 183 MOD 2

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for manufacture and Government acceptance of the Missile Container MK 183 MOD 2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

FED-STD-101	Preservation, Packaging and Packing Materials, Test Procedures

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Ordnance Systems Command

LD 541508	Container, Missile MK 183 MOD 2 and all documents listed thereon
DL 2642914	Sling, Weapons Handling MK 99 MOD 0
DL 2643551	Truck, Hand Lift MK 45 MOD 0

(Copies of Standards and specifications may be obtained from Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Drawings and other publications may be obtained from the procuring activity or as directed by the Contracting Officer).

FSC 8140

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3. REQUIREMENTS

3.1 General Requirements

3.1.1 Item Definition - The container MK 183 MOD 2 is a cylindrical horizontal container of aluminum construction, consisting of an upper and lower shell weldment. Latches around the periphery provide for rapid securing or opening. The upper and lower shell weldments are provided with clearance to accommodate the rocket motor fins and the airframe fins. The suspension system consists of shock-mounted cast aluminum supports secured to the lower shell weldment. The missile assembly is securely clamped to these supports by mating upper straps. A breather valve, tank valve, and humidity indicator are installed in the upper shell weldment. Skids, forklift pockets, stacking provisions, roll bars, and sling hook rings are provided to facilitate handling. A gasket seal is provided to maintain a closure seal. The overall dimensions are 194" long X 28 3/4" wide X 32 1/2" high, and the container weighs approximately 600 pounds. The Missile Container, referred to as container throughout this specification, will be used for the shipment and/or storage of the ASROC missile in various configurations.

3.1.2 Materials - Unless otherwise specified in the contract, requisition, or order, all materials, purchased parts, and processes thereto entering into the final assembly of the container, shall conform to the detailed requirements indicated on the drawings and specifications listed in LD 541508.

3.1.3 Precedence - If any conflict occurs between applicable documents, the order of precedence listed below shall be observed:

a. Contract or Purchase Order

b. This Specification

c. Applicable Drawings of LD 541508

d. Referenced Documents of LD 541508

e. Other applicable documents referenced in 2.1 of this specification. (The contractor shall notify the procuring activity of each instance of conflicting, or apparently conflicting, requirements.)

3.1.4 Construction - The container shall be manufactured in accordance with LD 541508 and all documents referenced thereon.

3.1.5 Clean-up - Before and after final assembly, all parts and components shall be thoroughly clean of dirt and other foreign matter.

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3.1.6 Workmanship - The container shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness. The standards of workmanship exhibited in any approved first article sample, subject to any qualifications stated in the Government notice of approval, shall establish the requirements of workmanship for production containers where such workmanship is not specifically covered by applicable specifications.

3.2 Performance Requirements

3.2.1 Characteristics - The purpose of performance requirements is to provide assurance that containers are manufactured in accordance with LD 541508 and drawings listed thereon. Containers manufactured in accordance with LD 541508 and drawings listed thereon, have satisfactorily met performance requirements contained herein.

3.2.2 First Article Sample - Unless otherwise specified in the contract or order, a representative sample of one unit shall be manufactured using the methods and procedures proposed for the production lot. The unit shall be submitted for first article tests at a government activity designated by the contracting agency to determine compliance with the requirements of the contract, specifications, and drawings. The first article sample shall withstand the tests specified in 4.6 of this specification except 4.6.7.2 and 4.6.8.2, shall meet all of the requirements specified in 3.1 of this specification, and the examination requirements of 4.5.4 of this specification.

3.2.3 Leakage - The container shall not leak more than 0.5 pounds per square inch gage (psig) over a period of 30 minutes, when pressurized to 1.0 plus 0.5 minus 0.0 psig.

3.2.4 Shock and Vibration - The container, with a dummy missile assembly, as government furnished equipment, (or simulated contents of equal weight distribution enclosed gross weight 1700 +170 -0 lbs.), shall withstand the shock and vibration tests specified in 4.6.3 without damage to the container or the contents, and without shifting of contents. Dented skids are permissible, but cracked welds and failed items are not. Evidence of shifting of contents within the container (creating conditions that may cause damage during shipment or binding as the contents are removed from the container) are not permissible.

3.2.5 Stacking - The loaded container - gross weight 1700 +170 -0 lbs shall be subjected to a load equivalent to that imposed by the stacking of four loaded containers upon the base container. The load shall be imposed for a period of 12 hours, and upon removal of the load, there shall be no evidence of damage to the contents or the container, sufficient to impair the removal of the contents.

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3.2.6 Skid Strength - The container, with a dummy missile assembly, (or simulated contents of equal weight distribution enclosed - gross weight 1700 +170 -0 lbs) shall be dragged along a concrete floor for a distance of five feet parallel to the skid length, and for a distance of five feet perpendicular to the skid length. Any misalignment of the skids, evidence of bending, or failure in the skid attachment, shall be cause for rejection.

3.2.7 Forklift Pocket Strength - The container, with a dummy missile assembly, (or simulated contents of equal weight distribution enclosed - gross weight 1700 +170 -0 lbs) shall be lifted by a fork truck negotiating a 15% grade and a bump, such as a rail crossing or sliding door sill. Vehicle speed during the test, shall be 5 ± 1 mph. Evidence of damage to the forklift openings, or instability of the container during the test shall be cause for rejection.

3.2.8 Hoisting Strength

3.2.8.1 Hoisting Strength - Entire Sling Hook System - The container (uniformly overloaded to a total weight five times its normal loaded weight-gross weight 8500 +850 -0 lbs.) shall be suspended for five minutes from a MK 99 sling (government furnished equipment) or a four legged sling of sufficient capacity, attached to the four container sling hooks. Upon removal of the load, there shall be no evidence of permanent deformation or failure.

3.2.8.2 Hoisting Strength - Entire Sling Hook System - The container with a dummy missile assembly (or contents of equal weight distribution enclosed - gross weight 1700 +170 -0 lbs.) shall be suspended for one minute from a sling of sufficient capacity such as the MK 99 sling attached to one forward sling hook and the diagonally opposite (aft) sling hook. The test shall be repeated using the remaining set of container sling hooks. Upon removal of the load there shall be no evidence of permanent deformation or failure.

3.2.8.3 Hoisting Strength - Single Hook - The container with a dummy missile assembly (or contents of equal weight distribution enclosed - gross weight 1700 +170 -0 lbs.) shall be suspended for five minutes from a sling of sufficient capacity, attached to one container sling hook. The test shall be repeated using a second container sling hook. Upon removal of the load, there shall be no evidence of permanent deformation or failure.

3.2.9 End Lift Strength

3.2.9.1 End Lift Strength - For this test, the container shall be loaded to three times its normal loaded weight (gross weight 5100 + 510 -0 lbs.). The loaded container shall be lifted clear of the deck by two MK 45 Hand Lift Trucks, (government furnished equipment) and held in position for a period of one minute. Upon removal of the load, there shall be no evidence of permanent deformation or failure.

3.2.9.2 End Lift Strength - For this test, the container shall be loaded to two times its normal weight (gross weight 3400 +340 -0 lbs.). The loaded container shall be lifted clear of the deck, by two MK 45 Hand Lift Trucks, and held in position for a period of one minute. Upon removal of the load, there shall be no evidence of permanent deformation or failure.

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3.3 Documentation Requirements

3.3.1 Documentation Notice - No documentation is required unless specified in the contract or purchase order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, purchase order, or requisition, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own, or other facilities acceptable to the procuring activity for the performance of the inspection requirements specified herein. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection - The inspection requirement specified herein are classified as follows:

a. First Article Inspection

b. Product Verification Inspection

c. Quality Conformance Inspection

4.3 First Article Acceptance Criteria - Acceptance shall be contingent upon no defects in the sample. Failure of the first article sample to comply with these requirements will result in rejection of the sample. Further production by the contractor prior to approval of the contracting agency of the first article samples, shall be at the contractor's risk.

4.4 Test Equipment - The following items of test equipment are required to perform the tests in this specification.

4.4.1 Compressor or Pressurized Tank - A compressor or pressurized tank shall have the capacity to pressurize the container to 1.0 plus 0.5 minus 0.0 psig. A pressure gage having an accuracy of plus or minus 10 percent shall be used to indicate the pressure in the container.

4.4.2 Vibration Test Machine - Vibration tests shall be conducted by any vibration machine capable of meeting the conditions specified in 4.6.3.1. Means shall be provided for controlling the direction of vibration of the machine and for adjusting and measuring the frequencies and amplitudes of vibration.

4.4.2.1 Hoist - A MK 99 MOD 0 Hoist (Government furnished equipment) or a four legged sling of sufficient capacity.

4.4.2.2 Lifting Equipment - Two MK 45 MOD 0 Hand Lift Trucks (Government furnished equipment).

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4.4.2.3 Dummy Load - A dummy load to simulate an ASROC missile and shall be provided by the Government. Other weights necessary for overweight testing as specified in 3.2.5, 3.2.8.1, 3.2.9.1 and 3.2.9.2, shall be the responsibility of the activity performing the testing.

4.5 Sampling

4.5.1 First Article Inspection - The first article sample, after satisfactorily passing the examinations of 4.5.4 shall be subjected to the tests specified in 4.6 except 4.6.7.2 and 4.6.8.2. Acceptance shall be based on no defects in the sample.

4.5.2 Quality Conformance Inspection - Each container shall be examined and comply with the requirements of 4.5.4 and shall be subjected to the tests specified in 4.6.2, 4.6.7.2, and 4.6.8.2. Acceptance shall be based on full compliance with the requirements.

4.5.3 Product Verification Tests - From each lot of 30 or less a random sample of (1) one container shall be selected for product verification tests. Product verification tests shall consist of those tests specified in paragraphs 4.6.3.3, 4.6.3.4 and 4.6.9. Acceptance of the lot shall be based on full compliance of the sample with the requirements of this specification. Failure to meet these requirements shall require the contractor to perform a failure analysis and submit a written report to the procuring activity stating the necessary corrective actions and or other recommendations.

4.5.4 Examination - Each container shall be examined for compliance with the requirements of NAVORD LD 541508 and drawings and specifications listed thereon.

4.6 Test Methods

4.6.1 Test Conditions - Unless otherwise specified herein, the following tests shall be performed in an area having a dry, level concrete surface condition (drop type tests may be performed on a steel surface) at ambient temperature. Tests shall be performed in the order listed in this specification.

4.6.2 Leakage Test - The container shall be leak tested in accordance with the pneumatic pressure technique of Method 5009 of FED-STD-101. The pressure shall be held for a minimum of 30 minutes and shall meet the requirements of 3.2.3.

4.6.3 Shock and Vibration Tests

4.6.3.1 Low Order Repetitive Shock Test - This test shall be run prior to running other standard shock tests This test shall be run in accordance with FED-STD-101 Method 5019 with the following clarifications:

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a. If circular input motion is used, table frequency shall be adjusted to assure that one edge of the container leaves the table on each cycle by 3/16 inch.

b. If vertical linear input motion is used, all points of the container shall leave the table by 1/16 inch at some instant during each cycle.

c. If neither of the above are attainable with equipment at hand, the table shall be operated at 270 ± 2 rpm for the stipulated time period.

Upon completion of the test, the container shall meet the requirements of 3.2.4.

4.6.3.2 Vibration Fatigue Test - The container shall be suitably instrumented to discern input (at the container base) and response (at the item c.g.) acceleration and suitably fastened in its normal storage position to a sinusoidal vibration generator of appropriate capacity. The container shall then be cycled from 5 to 60 Hz in 15 minutes with a logarithmic sweep rate. Input acceleration amplitude shall be a consistent 1 g. As an alternate for use with reaction type machines, the search syllabus of Table A may be followed.

Table A

Alternate Resonant Sweep Test Syllabus

Frequency Band Hz	Input Double Amplitude, In.	Minimum Sweep Time in Band, Min.
5 - 15	0.125	6.0
16 - 25	0.040	3.0
26 - 35	0.030	2.5
36 - 60	0.020	3.5

Peak transmissibility and the associated frequency shall be determined from the accelerometer records.

Set the frequency of the vibration generator at the peak transmissibility frequency disclosed by the survey sweep and vibrate at that frequency for 15 minutes at 1 g input acceleration. Upon completion of the test, evaluate the isolator system for excessive heat damage, breakage and continued ability to function. Excessive looseness in cushioned packs, evaluated one hour after completion of tests, shall be cause of rejection.

Upon completion of the test, the container shall meet the requirements of 3.2.4.

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4.6.3.3 Rotational Drop Test - This test shall conform to FED-STD-101 Method 5005 except that drop heights shall be 18 inches. Upon completion the container shall meet the requirements of 3.2.4.

4.6.3.4 Impact Test - Use incline impact, FED-STD-101 Method 5023 (without optional timber), or pendulum impact, FED-STD-101 Method 5012. Impact velocities shall be 8 feet per second. Upon completion of test, the container shall meet the requirements of 3.2.4.

4.6.4 Stacking Test - The loaded container shall be subjected to a load equivalent to that imposed by the stacking of four loaded containers upon the base containers and shall be imposed for a period of 12 hours. Upon completion of the test the container shall meet the requirements of 3.2.5.

4.6.5 Skid Strength Test - The loaded container shall be dragged along a concrete floor for a distance of five feet parallel to the skid length, and for a distance of five feet perpendicular to the skid length. Upon completion of the test, the container shall meet the requirements of 3.2.6.

4.6.6 Forklift Pocket Strength Test - The loaded container shall be lifted by a fork truck, negotiating a 15 percent grade and a bump, such as a rail crossing or sliding door sill. Vehicle speed during the test shall be 5 ± 1 mph. Upon completion of the test, the container shall meet the requirements of 3.2.7.

4.6.7 Hoisting Strength Tests

4.6.7.1 Hoisting Strength Test - Entire Sling Hook System (First Article Sample Only) - The Container (uniformly overloaded to a total weight five times its normal loaded weight) shall be suspended for five minutes from a four legged sling of sufficient capacity, attached to the four container sling hooks. Upon completion of the test, the container shall meet the requirements of 3.2.8.1.

4.6.7.2 Hoisting Strength Test - Entire Sling Hook System (Quality Conformance) - The loaded container shall be suspended for one minute from a sling of sufficient capacity, attached to one forward sling hook and the diagonally opposite aft sling hook. The test shall be repeated using the remaining set of container sling hooks. Upon completion of the test, the container shall meet the requirements of 3.2.8.2.

4.6.7.3 Hoisting Strength Test - Single Hook (First Article Sample Only) - The loaded container shall be suspended for five minutes from a sling of sufficient capacity, attached to one container sling hook. The test shall be repeated using a second container sling hook. Upon completion of the test, the container shall meet the requirements of 3.2.8.3.

4.6.8 End Lift Strength Tests

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4.6.8.1 End Lift Strength Test - (First Article Sample only) - The container (uniformly overloaded to a total weight three times its normal loaded weight) shall be lifted clear of the deck by two MK 45 Hand Lift Trucks, and held in this position for a period of one minute. Upon completion of the test, the container shall meet the requirements of 3.2.9.1.

4.6.8.2 End Lift Strength Test - (Quality Conformance) - The container (uniformly overloaded to a total weight two times its normal loaded weight) shall be lifted clear of the deck by two MK 45 Hand Lift Trucks, and held in this position for a period of one minute. Upon completion of the test, the container shall meet the requirements of 3.2.9.2.

4.6.9 Leakage Test - The leakage test shall be repeated in accordance with the procedures specified in paragraph 4.6.2. Upon completion of the test, the container shall meet the requirements of 3.2.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

Not applicable.

5.2 Packing

5.2.1 Level C - Unless otherwise specified, packing for domestic shipment shall be in accordance with approved commercial packing methods, adequate to protect the container from damage and to assure delivery at destination. The method used shall be commensurate with shipping at the lowest practical commercial rates.

5.3 Marking

5.3.1 Special Markings - None, unless otherwise specified.

5.3.2 Normal Markings - In addition to the markings required by contract or order, containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended Use - The container is intended for use in both foreign and domestic shipment and storage of one ASROC missile assembly.

6.2 Ordering Data - Procedure documents must specify the following information:

a. Title, number, and date of this specification

b. Number of containers to be furnished

c. Any exceptions of this specifications

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d. Any other pertinent requirements not covered herein.

6.3 Test Reports - When first article or product verification tests are performed, three copies of the test reports shall be supplied to the procuring activity and Naval Ammunition Depot Earle, Naval Weapons Handling Laboratory. Supplier test reports shall include complete details of the nature of the test, test methods, test equipment, test results, and result analysis. The test report shall be complete, so that no back-up or additional documentation is required for an evaluation of the report. When testing is performed by the Government, a full report will be issued to the contractor, the procuring activity and the Naval Weapons Handling Laboratory.

6.4 Materials Certification - The supplier shall furnish certification that all materials are in accordance with the requirements of LD 541508.

Preparing Activity:
Navy - OS
Proj. No. 8140-N080

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-C-82611(OS) Container, Missile MI 183 Mod 2

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

S/N 0102-014-1801 C-25254

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